Faculty of Computing

University of Sri Jayewardenepura

Undergraduate Final Year Project –Guideline (2026)

***The provisions outlined in this document are valid only for Final Year Projects**

(Course Code CIS4998 | CSE4998 | CCS4998)

conducted within the academic years 2026 and 2027, unless otherwise revised by the FoC.

Intended Learning Outcomes:

At the end of the module, the student will be able to:

MLO1: Describe effectively and efficiently to a range of audiences using appropriate technologies.

MLO2: Apply self-evaluation of the process.

MLO3: Design, review, implement, test, and verify the solution, giving due consideration to professional, ethical, security, and social responsibility, legal issues about the solution, and industry norms and best practices.

Intended Learning Outcomes (ILOs) can be elaborate as,

On successful completion of the Final Year Individual Project, the student will be able to:

- Identify and define a real-world or research problem
- Apply appropriate theoretical and technical knowledge
- Design and implement a methodological solution
- Analyse and interpret results critically
- Communicate findings effectively (oral & written)
- Demonstrate independent learning, ethics, and professionalism

Process Guide

The Final Year Project is a **compulsory academic requirement (Credits: 8, Duration: 1 Year: Appendix I: Assessment Breakdown and Appendix II: Tentative Timeline)** that allows student to apply the knowledge and skills student has gained during student's degree. The project is carried out individually and follows a structured process to ensure academic quality and fairness. Following are steps of the Individual Project.

Stages of the Final year Research Project

Prerequisites: Research topic and Supervisors

Students are permitted to propose a preferred research topic and supervisors. However, the appointment of supervisors is **subject to Faculty approval**, and the final allocation will be made by the Faculty based on **project domain alignment, availability of resources, and the academic benefit of the student**.

The Faculty will assign an **Internal Supervisor** and a **Senior Supervisor** for each project. The Senior Supervisor may be either from the Faculty or from another Faculty within the University of Sri Jayewardenepura. In cases where the Senior Supervisor is from another Faculty, the project must still have an **Internal Supervisor appointed by the Faculty** to ensure proper coordination and compliance with Faculty requirements.

Students may collaborate with an **External Supervisor**, particularly for interdisciplinary or industry-based projects. However, such arrangements must be **approved by the Faculty**, and the details of the External Supervisor must be formally submitted by the student.

1. Proposal Submission

Initially, students are required to submit a project proposal. The purpose of the proposal is to demonstrate that research project idea is feasible, relevant, and academically sound. Following topics should be included in the proposal.

- The proposal should clearly describe:
- The project title
- The problem or research question
- The objectives of the project (Outcome based Objectives around five)
- A brief literature background
- The method/s (tools, data, experiments, or analysis)
- Significance of the research
- A timeline for completion

All students should submit their proposed project proposal on or before the deadline mentioned in the Student Learning System (LMS). (Refer: SJP_FoC_UG_F002_Thesis_Proporal_Form)

2. Decision of Proposal Approval

After submission, student proposal will be reviewed by the supervisor and/or evaluation panel. The decision will be communicated via LMS to the individual student after one week (7 days) from the proposal deadline.

At this stage:

- The proposal may be approved without changes
- Approved with minor or major revisions
- Or rejected, requiring a new submission

Only approved proposals can proceed to the next stages. Feedback provided should be carefully addressed.

3. Oral Presentation for Proposal

Once the proposal is **approved or conditionally approved**, the student is required to submit a **recorded video presentation** of the proposed project.

The presentation should typically cover the following components:

- Background of the problem
- Project objectives
- Proposed method/s
- Expected results and significance

The purpose of this presentation is to assess the student's **understanding of the project** and their **ability to communicate ideas clearly and effectively**. The presentation must be submitted as a **short video clip (maximum 2 minutes)**, demonstrating the project concept in a **clear and creative manner**. The video must be uploaded to the **Learning Management System (LMS)** on or before the specified deadline (**within two weeks from the proposal approval date**).

4. Question and Answer Session for Proposal [10 marks]

Following the recorded oral presentation submission, student will face a Q&A session conducted by the evaluation panel (Supervisor/s).

Student may be asked about:

- Why student has chosen the topic
- Feasibility of the method
- Ethical issues (if any)
- Expected challenges and limitations

This session evaluates student's critical thinking, preparation, and subject knowledge. The proposal evaluation will be conducted by the supervisor/s.

5. Reflective Journal (40 Marks)

Every student is required to maintain a **Reflective Journal**, which provides a detailed account of the ongoing progress of the research project. The journal serves as a record of **continuous development, self-learning, and professional growth** throughout the project duration.

The Reflective Journal (10 marks each reflective journal report) must be submitted to the supervisor **once every two months** with the relevant **thesis chapters (Appendix III: Thesis Chapters)** for evaluation. The total allocation for this component is **40 marks**. **(Refer: SJP_FoC_UG_F003_Reflective Journal)**

The Reflective Journal should include the following:

- Work completed during the reporting period

- Problems encountered and corresponding solutions
- New skills or knowledge acquired
- Records of meetings with the supervisor
- Plan of work for the subsequent period
- Correlation of research progress with relevant thesis chapters

Milestone-Based Evaluation of the Reflective Journal

The Reflective Journal is structured around four key milestones:

1. Requirement Identification
2. Methods
3. Design
4. Implementation and evaluation

The final marks will reflect both the **practical progress of the project** and its **alignment with the thesis development (SJP_FoC_UG_F001_Thesis_Student_Guideline).**

Recommended Tools and Practices

Students are encouraged to use **standardized notations** (e.g., UML diagrams) where applicable.

Version control and code management tools such as GitHub are recommended to track development progress.

Communication and collaboration tools such as Google Groups may be used to facilitate discussions and include feedback from **external supervisors or domain experts**, where applicable.

Regular meetings with the supervisor should be conducted **at least once a week**, either online or in person, to ensure continuous monitoring and timely feedback.

6. Viva-voce examination (50 Marks)

The Viva Voce Examination is the final oral assessment of the student's research project. Students are required to upload a 15-minute recorded research presentation prior to the examination. **Upon submission of the thesis and presentation, the student must appear for a 20-minute oral examination**, which includes both the presentation (if required for clarification) and a question-and-answer session. During the viva voce examination, the student is expected to:

- Present the completed research project in a clear and structured manner
- Demonstrate the developed system, model, or results (where applicable)
- Explain the research methods, design decisions, and key findings
- Respond to technical, conceptual, and critical questions posed by the examiners

The viva voce examination is intended to assess the student's:

- Overall understanding of the project

- Depth of technical and theoretical knowledge
- Originality and individual contribution
- Ability to critically evaluate their work
- Communication and presentation skills

Following the viva voce examination, students will receive **examiner feedback via the Learning Management System (LMS)**. Students are required to **address and incorporate all comments and suggested revisions** before submitting the final **hard-bound thesis**. In cases where specific recommendations cannot be incorporated, the student must provide a **written justification**, which must be **reviewed and endorsed by the supervisor**. (Refer: SJP_FoC_UG_F007_These_Correction_Form).

7. Final Thesis Submission

Student must submit the final hard bound copies (two copies), soft copy of the thesis and viva-voce and the corrections on or before the date decided by the faculty. Codes, Datasets, Test cases and other relevant objects must be submitted along with the copies. **Final copy of the thesis submission will be considered as the end state of the Individual Project.**

Student should pass Proposal Presentation, Reflective Journal, Thesis evaluation and Viva Voce to be eligible obtain credits for the final year project.

Students are encouraged to submit their work for any reliable publication/ copy rights/ patent with the permission of the supervisor.

Other considerations

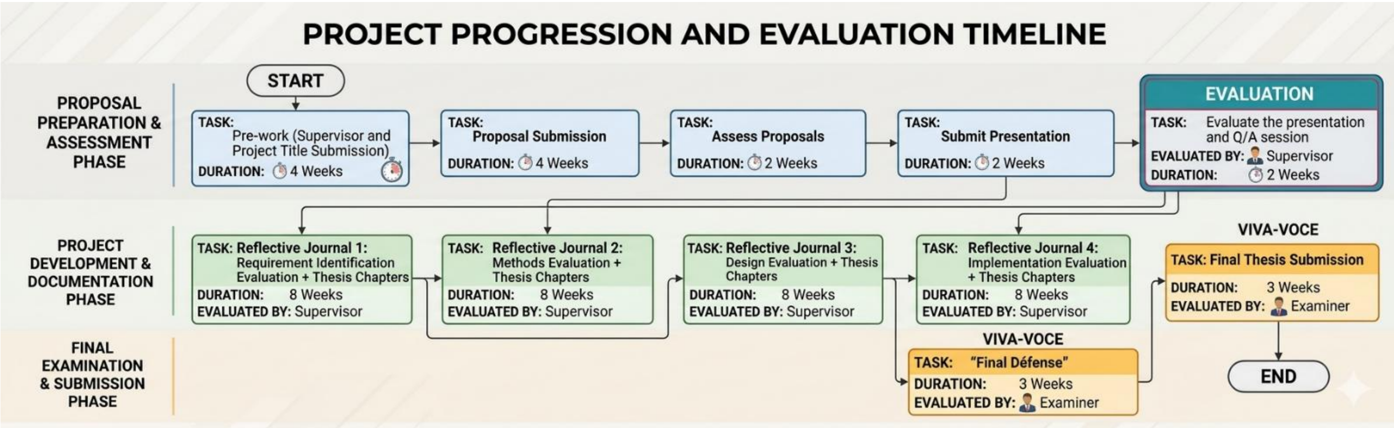
- Academic **honesty is mandatory** at all stages of the project.
- Students must **adhere to ethical research practices** throughout the project lifecycle.
- Students are **responsible for safe and ethical** data handling.
- Students may use AI tools **responsibly and transparently**.
- Respectful communication with supervisors, peers, and evaluators is expected.
- Supervisors provide guidance but **the work remains the student's responsibility.**
- If a student has concerns relating to an academic/personal relationship and is concerned that it is impacting on his/her studies, s/he should discuss this matter at his/her earliest opportunity, in confidence, with the Head of Department and Research coordinators.
- In case of changing the project topic (after the approval), students should inform the examination department at their earliest opportunity in written format along with the supervisor approval.
- In case of changing the supervisor, students should inform the examination department at their earliest opportunity in written format.

Appendix I: Assessment Breakdown

Component	**Evaluated by**	**Marks**
Proposal Submission	Supervisor	10
Reflective Journal	Supervisor	40
VIVA (Final Oral examination)	Examiner	50
Total		**100 Marks**

Appendix II: Tentative Timetable

ID	Task	Weeks	Start	Tentative Deadline
1	Pre-work (Supervisor and Project title submission)	4	20-Mar-2026	20-Apr-2026
2	Proposal Submission	4	20-Apr-2026	18-May-2026
3	Assess Proposals/ Assign Supervisors	2	18-May-2026	1-Jun-2026
4	Submit Presentation	2	1-Jun-2026	15-Jun-2026
5	Evaluate the presentation and Q/A session	2	15-Jun-2026	29-Jun-2026
6	Reflective Journal 1	8	29-Jun-2026	28-Aug-2026
7	Reflective Journal 2	8	28-Aug-2026	23-Oct-2026
8	Reflective Journal 3	8	23-Oct-2026	21-Dec-2026
9	Reflective Journal 4	8	21-Dec-2026	21-Feb-2027
12	Final Défense	3	21-Feb-2027	12-Mar-2027
13	Final Thesis Submission	2	12-Mar-2027	26-Mar-2027
14	Issue Results	1	26-Mar-2027	2-Apr-2027



Appendix III: Thesis Chapters

The following outline provides a general guideline for the structure of the thesis and indicates the expected content to be addressed in each chapter. The listed sections are not intended to be mandatory subsections; students may adapt and incorporate them as appropriate, based on the relevance to their research. Each chapter should be written concisely and clearly. As a general guideline, the length of a chapter should be approximately **10 pages**, unless otherwise approved by the supervisor based on the nature of the research.

1. Introduction Chapter
    - Background
    - Research Question
    - Outcome Oriented SMART Objectives/Hypothesis (*Need to trace back from conclusion)
    - Purpose & Significance
2. Literature Review Chapter
    - Relevant Theoretical Foundations
    - Discipline Specific Literature review
    - Application Area/Business Sector Specific Literature review
    - Similar Adoption from different Sectors/Domains
    - Introduction of (On-going) Case Study
3. Research Method/s Chapter
    - Methodological Foundation (SRM | DSRM | AR | GT | BS | SNA)
    - Qualitative / Quantitative Methods
    - Facts Finding Techniques (Interviews, Questionnaires, etc.)
    - Research Process/Approach Phases of the research approach/es as instantiated for specific project
    - Procedures to adopt to meet objective/s
4. Design Chapter
    - Design diagrams with Standard Notations and Specifications (Ex:UML, MPMN, 4+1 View, ISO)
    - As-is Design (Domain Model)
    - System Architecture
    - To-be Design (System Model)
    - Functional View, Static View, Dynamic View
    - Elaboration phase of user perspective
    - Proposed/Modified Algorithms
5. Implementation / Application Chapter
    - Deployment Diagrams

- GUIs
- Elaboration on last two phases of User Perspectives
- The PoC Implementations

6. Evaluation Chapter
    - Criteria: Suitability, Generalizability, Reliability and Validity of Contributions
    - Empirical Evaluation
    - Theoretical/Laboratory Evaluation
    - Stakeholder/User Feedbacks
    - Performance Measures
7. Conclusion and Future Work Chapter
    - Achievements of Objectives [Refer as per formulations in introduction]
    - Acceptance/Rejection of the set hypothesis
    - Research Claim as per set Objectives
    - Reasons (Theories/Logics) for the Claims
    - Evidence (Primary/Secondary data)
    - Further Work/Future Directions (to overcome limitations)
8. References
    - Relevance
    - References to state-of-the-art Knowledge
    - Types of sources
    - Coverage of varieties
    - Tool utilizations
9. Annexures
10. Program Listings
11. Screen Shots
12. Surveys/Feedback

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